

FEAR ITSELF IS POWERFUL

NONCONSUMPTIVE EFFECTS OF PREDATORS ON THEIR PREY

IN CONTEXT

KEY FIGURE

Earl Werner (1944–)

BEFORE

1966 American ecologist Robert Paine conducts a series of groundbreaking field experiments to highlight the crucial effects of a predator on the community in which it lives.

1990 Canadian biologists Steven Lima and Lawrence Dill analyzed the decision-making of organisms that are at the greatest risk of being preyed on by other creatures.

AFTER

2008 American behavioral biologist and ecologist John Orrock teams up with Earl Werner and others to produce mathematical models to explain the nonconsumptive effects of predatory animals.

Many descriptions of ecosystems focus on predator–prey interactions in which predators kill and prey are eaten. However, American ecologist Earl Werner and others have shown that the mere presence of a predator affects the behavior of prey.